GlucoTwin

An India-aware digital twin for proactive type 2 diabetes management

Team ZIZ, Chandigarh University

Happiest Health Digital Twin Challenge 2026



Synthetic, open or anonymized data only

Decision support, not a medical device

Diabetes care is mostly reactive

Seen every few months

Patients meet the doctor only every few months.

Spikes noticed afterwards

A rise in glucose is usually seen after it happened.

No continuous view

Between visits the doctor sees nothing continuous.

Our goal: give the doctor a continuous view between visits, and warn about a likely excursion before it happens, not after.

What a digital twin is

A virtual copy of one patient, kept up to date with live data, that can monitor, predict and answer what-if questions.

A normal prediction model

- ✗ Not personalized per patient
- ✗ Not continuously updated
- ✗ No what-if testing

GlucoTwin digital twin

- ✓ Personalized per patient
- ✓ Continuously updated
- ✓ Supports what-if testing

GlucoTwin at a glance



Monitor

Live replay of the CGM stream



Predict

Glucose at +30, +60 and +120 min with a q10 to q90 band



Alert

Only events that pass a safety gate



Explain

Top features behind each forecast



Personalize

Small per-patient adapters



What-if

Change a meal or activity and re-run



Doctor dashboard

Five pages for the clinician

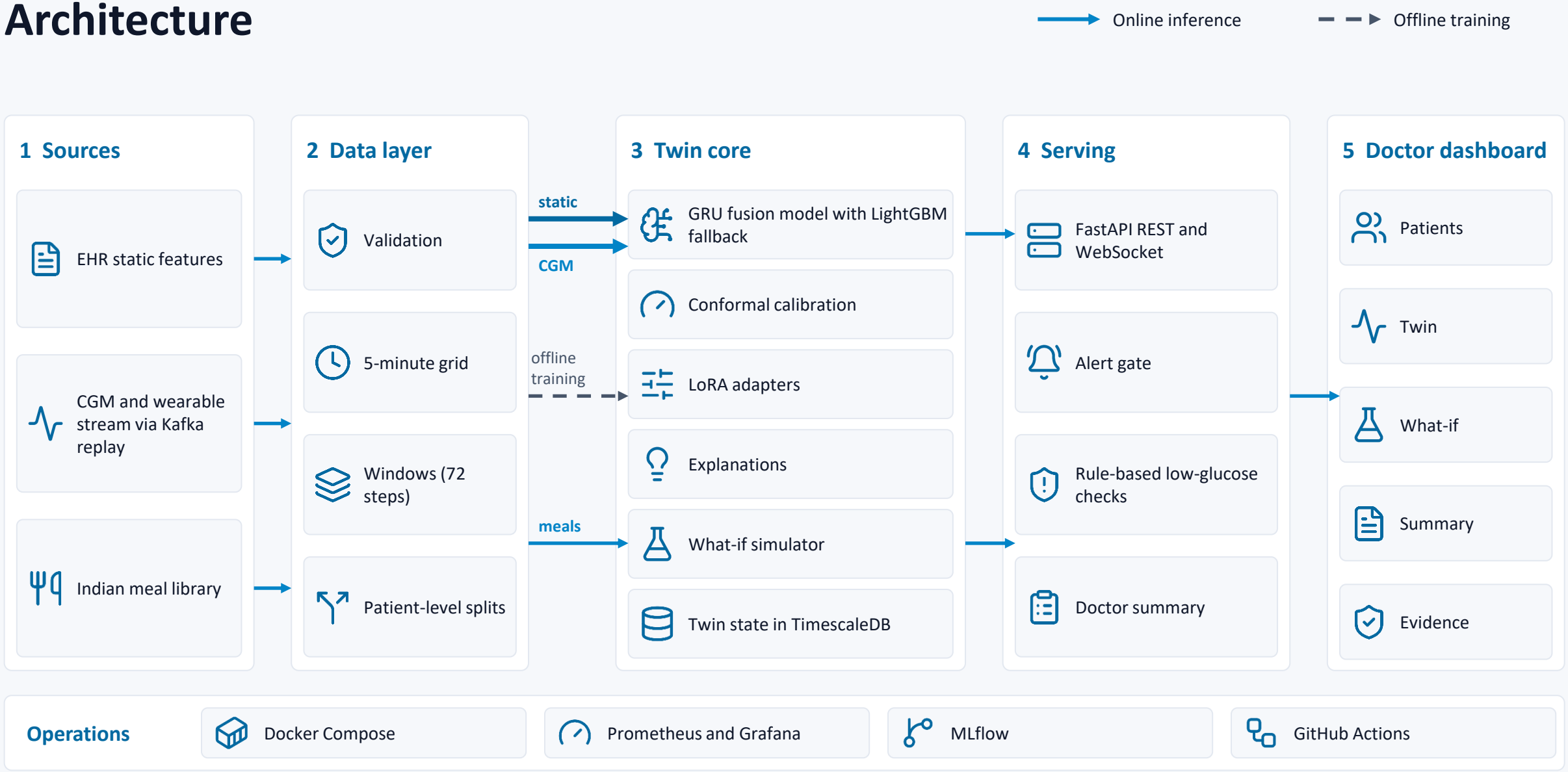
Data sources, roles and licenses

Source	Role	Size and notes	License
CGMacros (PhysioNet)	All training and evaluation	45 participants (14 type 2 diabetes, 16 prediabetes, 15 healthy); 134,292 prediction windows	CC BY-NC-SA 4.0
HUPA-UCM (Mendeley Data)	Optional pretraining only, never used for evaluation	Type 1, 25 patients	CC BY 4.0
Synthetic AI-READI (ADVANCE Center)	Static patient table only	10,518 synthetic static records	Terms not published
Indian meal library	What-if meals	20 dishes; team estimates from IFCT 2017 raw ingredient values plus recipe assumptions; glycemic index values unverified	Team-made

No raw data redistributed

Record-to-wearable link is synthetic

Architecture



How the model fuses two streams

Static features

Health record data, turned into a static embedding

CGM sequence

5-minute grid, 6-hour lookback (72 steps). Known-future inputs: hour of day, time since last meal, recent carbs.



Outputs

- Forecasts at +30, +60 and +120 min
- Median plus q10 to q90 band
- Conformal calibration, 80% target

80.2%

of observed values fell inside the GRU interval at +30 min (80% target). Static and time-varying features are kept separate.

Evaluation design: no patient leaks across folds

5-fold leave-patients-out cross-validation

Stratified by diabetes status

Scalers fit on training patients only

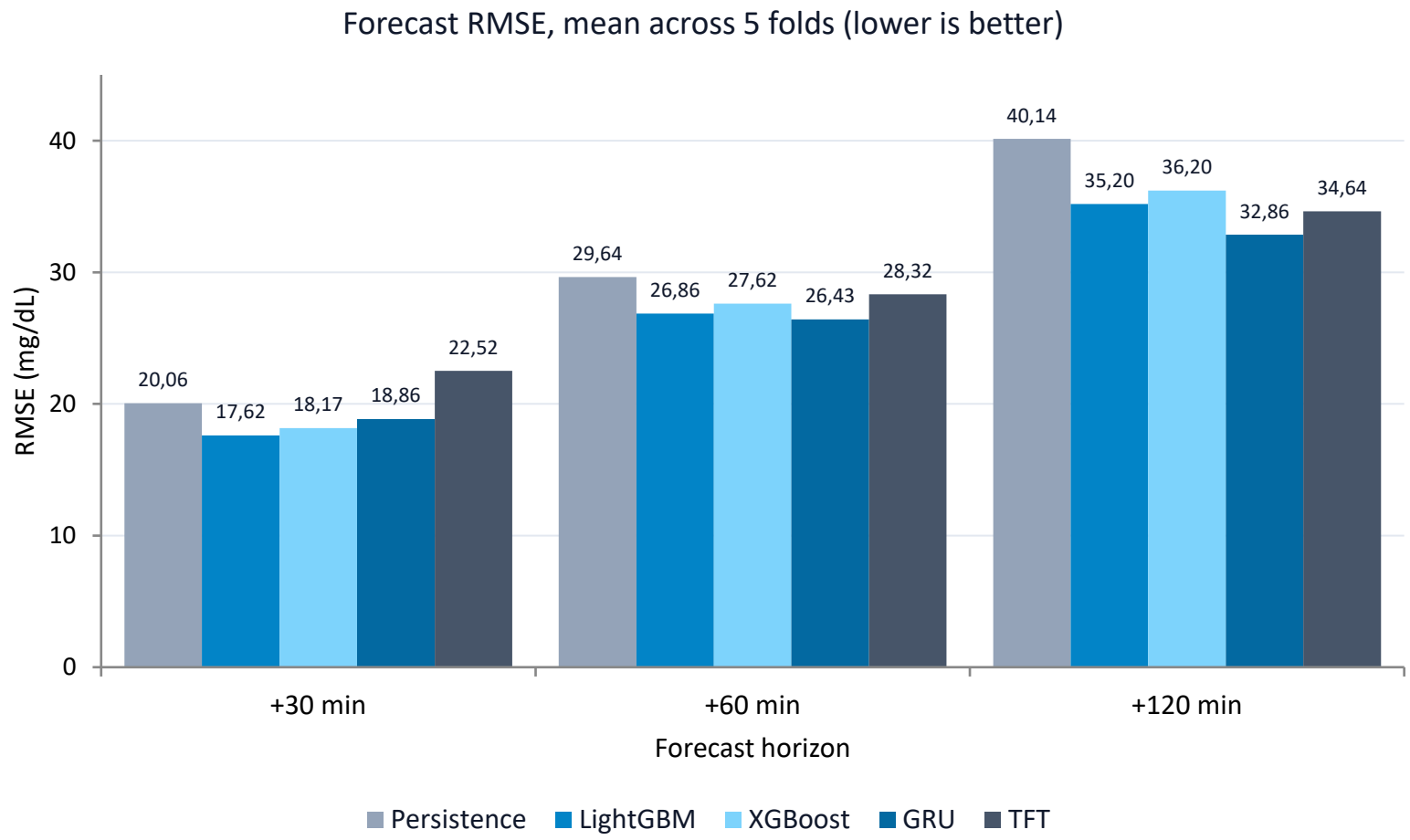
No patient in more than one role

Events we predict

- **Hyperglycemia: above 180 mg/dL**
- **Severe: above 250 mg/dL**
- **Hypoglycemia: below 70 mg/dL**
- **Spike: rise of 40 mg/dL or more within 60 minutes**

134,292 prediction windows from 45 participants were used for all training and evaluation.

Forecast error by horizon



32.86

mg/dL, GRU at +120 min, against 40.14 for persistence

A modest improvement over persistence. At +30 and +60 min the GRU, LightGBM and XGBoost are within one standard deviation of each other.

Paired comparison with persistence

Mean reduction in per-patient RMSE over 45 patients, with 95% bootstrap interval

Horizon	Model	Mean reduction (mg/dL)	95% interval	Patients improved
+30 min	LightGBM	2.39	[1.92, 2.83]	43 of 45
+30 min	XGBoost	1.99	[1.35, 2.54]	41 of 45
+30 min	GRU	1.41	[0.83, 1.95]	36 of 45
+60 min	GRU	3.21	[2.09, 4.26]	36 of 45
+60 min	LightGBM	2.85	[1.85, 3.78]	39 of 45
+60 min	XGBoost	2.36	[1.16, 3.47]	37 of 45

The intervals exclude zero, but the reductions are modest, between 1.41 and 3.21 mg/dL, and not every patient improves.

Alerts only when a safety gate is passed

The alert gate

- Held-out recall at least 40%
- Precision at least 30%
- Reliable: at least 10 events per fold
- At most 1.0 false alerts per patient-day

Hyperglycemia at +30 min

65 ± 10%

recall

61 ± 4%

precision

0.8 ± 0.2

false alerts per patient-day

18 min

median lead time

Five events pass

Hyperglycemia +30 min. Severe hyperglycemia +30 and +60 min. Spike +30 and +60 min. All other events are shown only as a risk estimate, never an alert.

Low glucose: rule-based, not predicted

Predicted hypoglycemia did not pass (too few events). Checks on the observed reading: below 70, below 54, rapid fall.

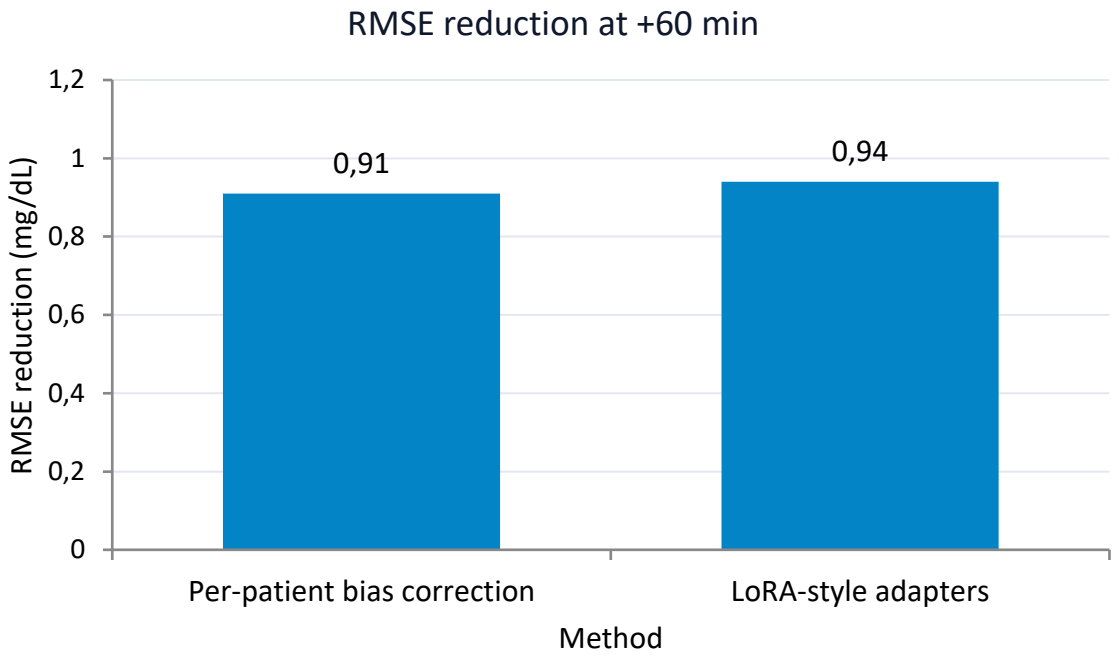
Explanations and personalization

Explanations

Permutation importance for the GRU, SHAP for LightGBM.

39 of 45

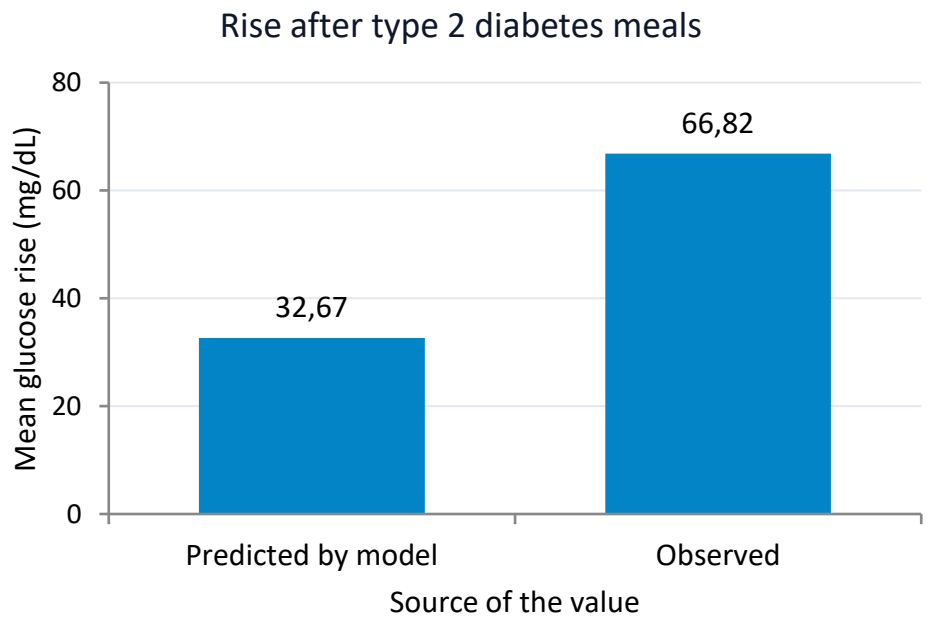
patients improved with per-patient adapters. MARD 12.82% to 11.90%. RMSE 24.14 to 23.20 mg/dL at +60 min.



A small effect. A simple per-patient bias correction gives almost the same RMSE gain (0.91 vs 0.94 mg/dL). Adapters are promoted only if held-out error improves.

What-if simulator

Re-runs the model with a changed meal or activity. Default is the raw model output.



98.4%

of meals show a positive rise (direction is right)

0.3994 ± 0.1102

cross-fold Pearson r

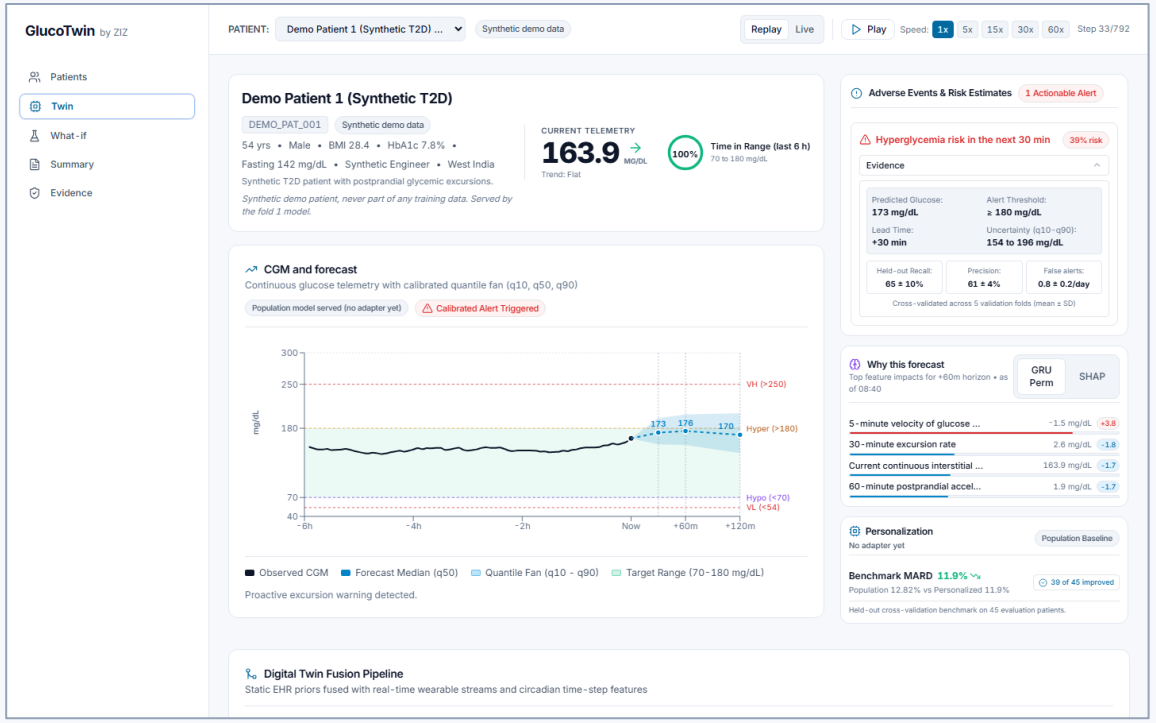
72.5% to 100%

monotonic in carbohydrate: raw model, then physiological projection (100% by construction)

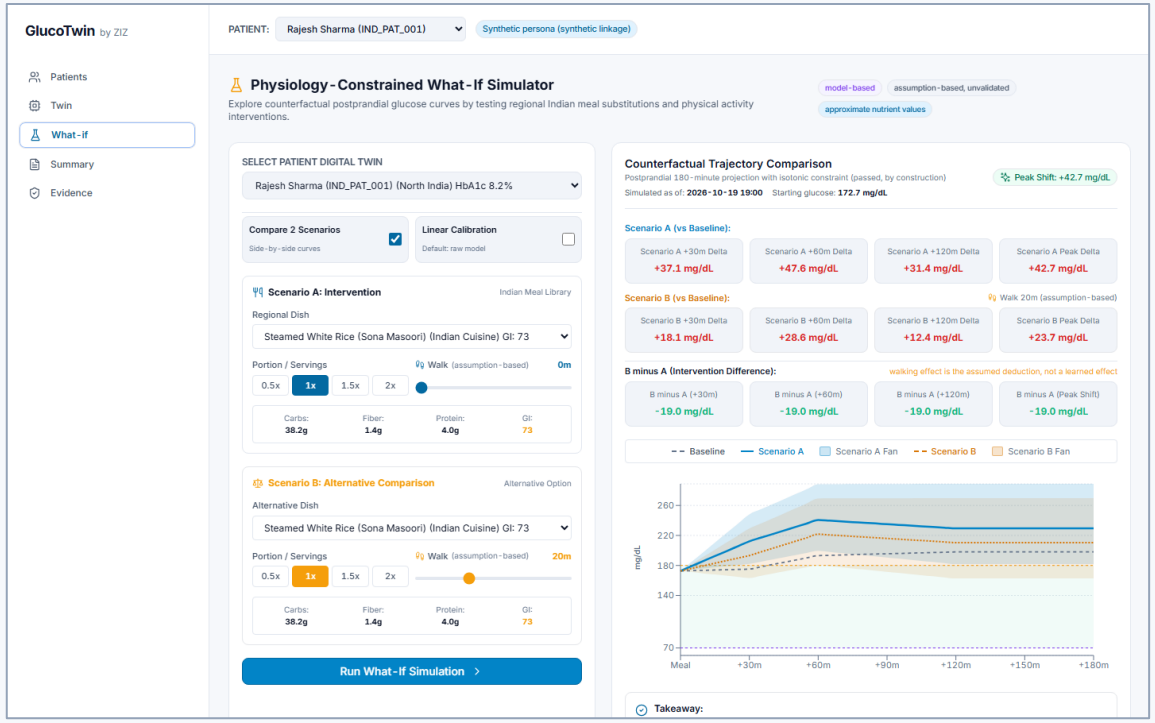
Meal rises are underestimated by about half. The walking effect is an assumption (0.38 mg/dL per net MET-minute, capped at 50 mg/dL), not validated.

model-based | assumption-based | approximate nutrient values

Doctor dashboard



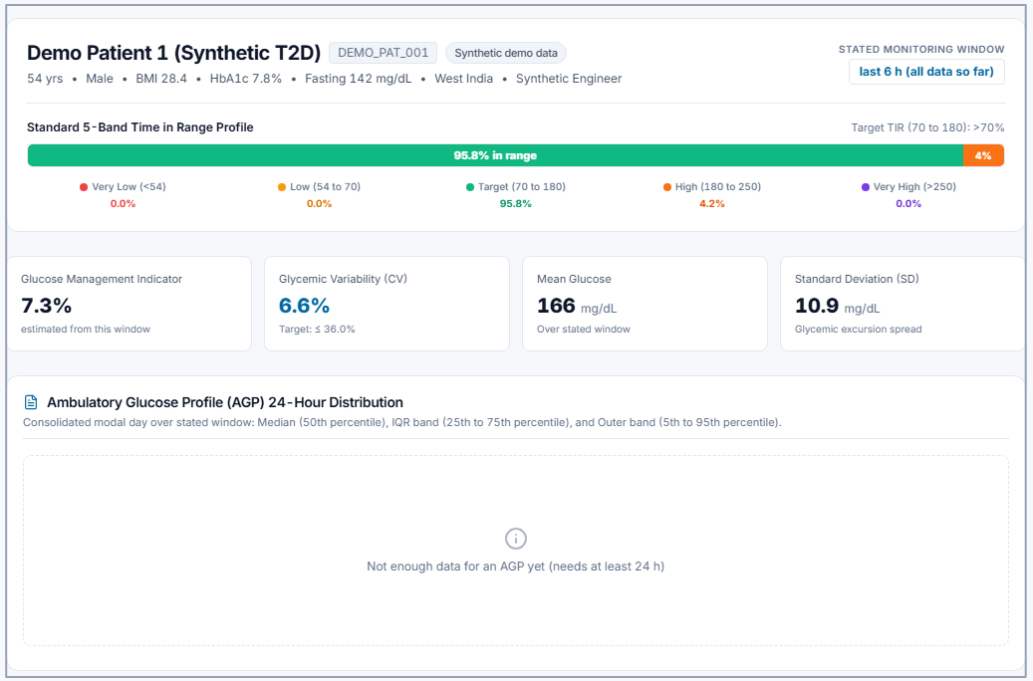
Twin page: replay, forecast fan, alert, explanation, personalization



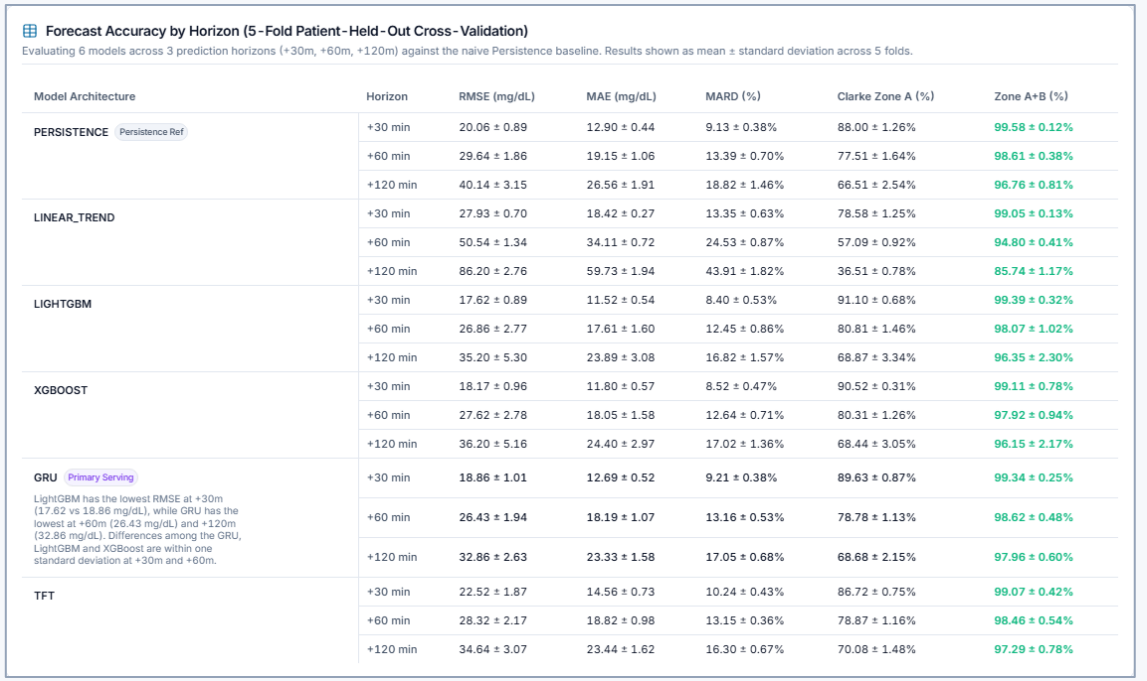
What-if page: two scenarios against a baseline, with provenance badges

Five pages: Patients, Twin, What-if, Summary, Evidence. The demo uses 3 synthetic demo patients and 4 synthetic personas.

Summary and Evidence pages



Summary page: time in range, GMI and CV. The AGP appears when at least 24 h of data exist.



Forecast Accuracy by Horizon (5-Fold Patient-Held-Out Cross-Validation)

Evaluating 6 models across 3 prediction horizons (+30m, +60m, +120m) against the naive Persistence baseline. Results shown as mean ± standard deviation across 5 folds.

Model Architecture	Horizon	RMSE (mg/dL)	MAE (mg/dL)	MARD (%)	Clarke Zone A (%)	Zone A+B (%)
PERSISTENCE (Persistence Ref)	+30 min	20.06 ± 0.89	12.90 ± 0.44	9.13 ± 0.38%	88.00 ± 1.26%	99.58 ± 0.12%
	+60 min	29.64 ± 1.86	19.15 ± 1.06	13.39 ± 0.70%	77.51 ± 1.64%	98.61 ± 0.38%
	+120 min	40.14 ± 3.15	26.56 ± 1.91	18.82 ± 1.46%	66.51 ± 2.54%	96.76 ± 0.81%
LINEAR_TREND	+30 min	27.93 ± 0.70	18.42 ± 0.27	13.35 ± 0.63%	78.58 ± 1.25%	99.05 ± 0.13%
	+60 min	50.54 ± 1.34	34.11 ± 0.72	24.53 ± 0.87%	57.09 ± 0.92%	94.80 ± 0.41%
	+120 min	86.20 ± 2.76	59.73 ± 1.94	43.91 ± 1.82%	36.51 ± 0.78%	85.74 ± 1.17%
LIGHTGBM	+30 min	17.62 ± 0.89	11.52 ± 0.54	8.40 ± 0.53%	91.10 ± 0.68%	99.39 ± 0.32%
	+60 min	26.86 ± 2.77	17.61 ± 1.60	12.45 ± 0.86%	80.81 ± 1.46%	98.07 ± 1.02%
	+120 min	35.20 ± 5.30	23.89 ± 3.08	16.82 ± 1.57%	68.87 ± 3.34%	96.35 ± 2.30%
XGBOOST	+30 min	18.17 ± 0.96	11.80 ± 0.57	8.52 ± 0.47%	90.52 ± 0.31%	99.11 ± 0.78%
	+60 min	27.62 ± 2.78	18.05 ± 1.58	12.64 ± 0.71%	80.31 ± 1.26%	97.92 ± 0.94%
	+120 min	36.20 ± 5.16	24.40 ± 2.97	17.02 ± 1.36%	68.44 ± 3.05%	96.15 ± 2.17%
GRU (Primary Serving)	+30 min	18.86 ± 1.01	12.69 ± 0.52	9.21 ± 0.38%	89.63 ± 0.87%	99.34 ± 0.25%
	+60 min	26.43 ± 1.94	18.19 ± 1.07	13.16 ± 0.53%	78.78 ± 1.13%	98.62 ± 0.48%
	+120 min	32.86 ± 2.63	23.33 ± 1.58	17.05 ± 0.68%	68.68 ± 2.15%	97.96 ± 0.60%
TFT	+30 min	22.52 ± 1.87	14.56 ± 0.73	10.24 ± 0.43%	86.72 ± 0.75%	99.07 ± 0.42%
	+60 min	28.32 ± 2.17	18.82 ± 0.98	13.15 ± 0.36%	78.87 ± 1.16%	98.46 ± 0.54%
	+120 min	34.64 ± 3.07	23.44 ± 1.62	16.30 ± 0.67%	70.08 ± 1.48%	97.29 ± 0.78%

LightGBM has the lowest RMSE at +30m (17.62 vs 18.86 mg/dL), while GRU has the lowest at +60m (26.43 mg/dL) and +120m (32.86 mg/dL). Differences among the GRU, LightGBM and XGBoost are within one standard deviation at +30m and +60m.

Evidence page: every number is read from the saved reports.

The Evidence page lets a reviewer check the claims in this deck against the saved reports.

Engineering and reproducibility

125

backend tests

37

frontend tests

about 4.6 ms

median inference latency

about 34 to 44 ms

median end-to-end stream latency, demo stack

FastAPI

REST and WebSocket

Kafka

Replay of the wearable stream

PostgreSQL

with TimescaleDB

Prometheus and Grafana

Monitoring

MLflow

Experiment tracking

Docker Compose

One command starts everything; backend image 2.37 GB

GitHub Actions

4 jobs: lint and types, backend tests, frontend build, Docker build

Licenses

Code MIT; models and reports CC BY-NC-SA 4.0

No raw data is included. Built with AI-assisted coding (Google Antigravity); the team specified, reviewed and tested everything.

Limitations and safety

Limitations

- Only 14 type 2 diabetes participants
- Record-to-wearable link is synthetic
- Hypoglycemia prediction is unreliable
- Meal rises underestimated by about half
- Meal nutrient values are estimates; glycemic index values unverified
- Activity effect is an assumption
- No prospective or clinical study

Safety design

- Decision support, not a medical device
- No clinical validation, diagnosis or regulatory status is claimed
- Only five events can raise an alert
- Low-glucose checks are labeled rule-based, not predicted
- Every input shows a provenance badge

Impact and roadmap

What it offers

- A continuous view for the doctor between visits
- Predicted warnings for five event types instead of noticing spikes afterwards
- Indian meals for what-if questions, as estimates

Roadmap

- Validate on larger Indian cohorts
- Device and FHIR integration
- Clinician feedback and usability testing
- Better hypoglycemia detection
- A prospective pilot
- Guideline-grounded explanations with citations (future work, not built)
- Photo-based meal logging (future work)

Thank you

Team ZIZ, Chandigarh University

Happiest Health Digital Twin Challenge 2026

Team Leader: Likhith Sai Seemakurthi
Member: Pritesh Kumar Pandey

GitHub repository

priteshkumarpandey/ZIZ_CHANDIGARH-UNIVERSITY

Demo video

[VIDEO LINK] placeholder

A1 Full RMSE table

Forecast RMSE in mg/dL, mean ± SD across 5 folds. Lowest in each column in blue.

Model	+30 min	+60 min	+120 min
Persistence	20.06 ± 0.89	29.64 ± 1.86	40.14 ± 3.15
LightGBM	**17.62 ± 0.89**	26.86 ± 2.77	35.20 ± 5.30
XGBoost	18.17 ± 0.96	27.62 ± 2.78	36.20 ± 5.16
GRU (served primary)	18.86 ± 1.01	**26.43 ± 1.94**	**32.86 ± 2.63**
TFT (experiment)	22.52 ± 1.87	28.32 ± 2.17	34.64 ± 3.07

Clarke error grid zones A+B at +30 min: LightGBM 99.39 ± 0.32 percent, GRU 99.34 ± 0.25 percent.

A2 Alert operating table

Gate: recall at least 40%, precision at least 30%, reliable (at least 10 events per fold), at most 1.0 false alerts per patient-day.

Event	Status	Operating point
Hyperglycemia +30 min	Alert	Recall 65 ± 10%, precision 61 ± 4%, 0.8 ± 0.2 false alerts per patient-day, median lead time 18 min
Severe hyperglycemia +30 min	Alert	See Evidence page
Severe hyperglycemia +60 min	Alert	See Evidence page
Spike +30 min	Alert	See Evidence page
Spike +60 min	Alert	See Evidence page
All other events	Risk estimate only	Not shown as alerts
Predicted hypoglycemia	Did not pass (too few events)	Rule-based checks on observed reading: below 70, below 54, rapid fall

A3 Data and license table

Source or artifact	Role	License	Raw data redistributed
CGMacros (PhysioNet)	Training and evaluation	CC BY-NC-SA 4.0	No
HUPA-UCM (Mendeley Data)	Optional pretraining only	CC BY 4.0	No
Synthetic AI-READI (ADVANCE Center)	Static patient table only	Terms not published	No
Indian meal library	What-if meals (team estimates, IFCT 2017 based)	Team-made	No
Code	GlucoTwin software	MIT	Not applicable
Trained model files and reports	Served models and evidence	CC BY-NC-SA 4.0 (non-commercial, attribution, share-alike)	No raw data included

A4 Why not an LLM or RAG?

We did not build one. The task is numeric forecasting of a glucose time series with uncertainty, and our models can be tested on held-out patients. Our explanations come from permutation importance and SHAP, which point to model inputs. Guideline-grounded explanations with citations are future work and are not built.

A5 Why a GRU and not a transformer?

We tried a Temporal Fusion Transformer as an experiment. At +30 and +60 min it had the highest RMSE of the learned models (22.52 and 28.32 mg/dL), and at +120 min the GRU was lower (32.86 against 34.64). The GRU is lowest at +60 and +120 min and fuses the static embedding with the CGM sequence. LightGBM is lowest at +30 min, within one standard deviation, so it is the fallback.

A6 How do you know it generalizes?

We only know it partly. Evaluation uses 5-fold leave-patients-out cross-validation, stratified by diabetes status, with scalers fit on training patients only, and paired comparisons with bootstrap intervals over 45 patients. But there are only 45 participants, 14 with type 2 diabetes, the record-to-wearable link is synthetic, and there is no prospective or clinical study.

A7 Is hypoglycemia handled?

Not by prediction. Predicted hypoglycemia did not pass the alert gate because there were too few events, so it is shown only as a risk estimate. Low glucose is covered by rule-based checks on the observed reading (below 70, below 54, rapid fall), labeled rule-based, not predicted. Better hypoglycemia detection is on the roadmap.

A8 What are the main limits?

Only 14 type 2 diabetes participants; a synthetic link between record and wearable; unreliable hypoglycemia prediction; meal rises underestimated by about half; estimated meal nutrient values and unverified glycemic index values; an activity effect that is an assumption; and no prospective or clinical study.